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DEVELOPMENT OF THE DIGITAL GAME "BORA CATÁ" TO SUPPORT AWARENESS OF SELECTIVE WASTE COLLECTION

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ABSTRACT

Objective: The objective of this study is to develop and evaluate the digital game "BORA CATÁ" as a tool to promote awareness about selective waste collection and its impact on sustainable practices.

Theoretical Framework: The study draws on concepts from environmental education, serious games, and the role of technology in behavior change. It aligns with the Sustainable Development Goals (SDGs), focusing on education and environmental responsibility.

Method: The research employed a design and development methodology for the creation of the game using the Unity platform. The study included iterative testing phases with feedback collection from users to refine gameplay and educational content. Participants' perceptions were gathered through surveys to evaluate engagement, usability, and learning outcomes.

Results and Discussion: Initial feedback highlighted the game's potential to engage users and improve their understanding of proper waste management. Players reported increased awareness of environmental impacts and expressed a preference for learning through interactive and enjoyable methods. The study also identified areas for improvement, including interface accessibility and age-appropriate adjustments.

Research Implications: This research contributes to the field by demonstrating how persuasive games can bridge entertainment and education to foster behavioral change. The findings support the integration of digital tools into environmental education strategies.

Originality/Value: This study innovates by addressing waste collection awareness through an interactive digital platform, emphasizing the potential of serious games to influence sustainable practices and support global environmental goals.

Keywords: Serious Games, Recycling, Environmental Education, Sustainable Behavior, Persuasive Games.

DESENVOLVIMENTO DO JOGO DIGITAL "BORA CATÁ" PARA APOIAR A SENSIBILIZAÇÃO À COLETA SELETIVA DE RESÍDUOS

RESUMO

Objetivo: O objetivo deste estudo é desenvolver e avaliar o jogo digital "BORA CATÁ" como uma ferramenta para promover a conscientização sobre a coleta seletiva e seu impacto em práticas sustentáveis.

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Referencial Teórico: O estudo baseia-se em conceitos de educação ambiental, jogos sérios e o papel da tecnologia na mudança de comportamento. Ele está alinhado aos Objetivos de Desenvolvimento Sustentável (ODS), com foco na educação e responsabilidade ambiental.

Método: A pesquisa utilizou uma metodologia de design e desenvolvimento para a criação do jogo usando a plataforma Unity. O estudo incluiu fases iterativas de testes com a coleta de feedback dos usuários para refinar a jogabilidade e o conteúdo educativo. As percepções dos participantes foram coletadas por meio de questionários para avaliar engajamento, usabilidade e resultados de aprendizagem.

Resultados e Discussão: Os feedbacks iniciais destacaram o potencial do jogo para engajar os usuários e melhorar sua compreensão sobre o gerenciamento adequado de resíduos. Os jogadores relataram maior conscientização sobre os impactos ambientais e expressaram preferência por aprender de forma interativa e agradável. O estudo também identificou áreas para melhorias, como acessibilidade da interface e ajustes apropriados para diferentes faixas etárias.

Implicações da Pesquisa: Esta pesquisa contribui para a área ao demonstrar como jogos persuasivos podem unir entretenimento e educação para promover mudanças comportamentais. Os resultados apoiam a integração de ferramentas digitais em estratégias de educação ambiental.

Originalidade/Valor: Este estudo inova ao abordar a conscientização sobre coleta seletiva por meio de uma plataforma digital interativa, destacando o potencial dos jogos sérios para influenciar práticas sustentáveis e apoiar metas ambientais globais.

Palavras-chave: Jogos Sérios, Reciclagem, Educação Ambiental, Comportamento Sustentável, Jogos Persuasivos.

DESARROLLO DEL JUEGO DIGITAL “BORA CATÁ” PARA APOYAR LA SENSIBILIZACIÓN DE LA RECOGIDA SELECTIVA DE RESIDUOS

RESUMEN

Objetivo: El objetivo de este estudio es desarrollar y evaluar el juego digital "BORA CATÁ" como una herramienta para promover la conciencia sobre la recolección selectiva y su impacto en las prácticas sostenibles.

Marco Teórico: El estudio se basa en conceptos de educación ambiental, juegos serios y el papel de la tecnología en el cambio de comportamiento. Se alinea con los Objetivos de Desarrollo Sostenible (ODS), con un enfoque en la educación y la responsabilidad ambiental.

Método: La investigación empleó una metodología de diseño y desarrollo para la creación del juego utilizando la plataforma Unity. El estudio incluyó fases iterativas de pruebas con la recopilación de comentarios de los usuarios para refinar la jugabilidad y el contenido educativo. Las percepciones de los participantes se recopilaron a través de encuestas para evaluar el compromiso, la usabilidad y los resultados de aprendizaje.

Resultados y Discusión: Los comentarios iniciales destacaron el potencial del juego para involucrar a los usuarios y mejorar su comprensión sobre la gestión adecuada de residuos. Los jugadores informaron un mayor nivel de conciencia sobre los impactos ambientales y expresaron una preferencia por aprender de manera interactiva y agradable. El estudio también identificó áreas de mejora, como la accesibilidad de la interfaz y ajustes adecuados para diferentes grupos de edad.

Implicaciones de la Investigación: Esta investigación contribuye al campo al demostrar cómo los juegos persuasivos pueden combinar entretenimiento y educación para fomentar cambios de comportamiento. Los resultados respaldan la integración de herramientas digitales en estrategias de educación ambiental.

Originalidad/Valor: Este estudio innova al abordar la conciencia sobre la recolección selectiva a través de una plataforma digital interactiva, destacando el potencial de los juegos serios para influir en prácticas sostenibles y apoyar objetivos ambientales globales.

Palabras clave: Juegos Serios, Reciclaje, Educación Ambiental, Comportamiento Sostenible, Juegos Persuasivos.

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1 INTRODUCTION

In contemporary society, environmental issues, particularly those related to waste management, represent critical challenges with global implications. Despite the availability of recyclable materials, a significant portion is improperly discarded, contributing to pollution of soil, water, and air. The inadequate management of solid waste has severe consequences, including the obstruction of waterways, the contamination of natural ecosystems, and negative impacts on public health and quality of life. These challenges demand innovative approaches to foster environmental awareness and responsible behavior.

Education plays a vital role in promoting sustainable practices and instilling ecological consciousness. As children represent the next generation of global citizens, their involvement in environmental education is crucial. Both family and educational institutions serve as foundational platforms for shaping sustainable habits. However, with the increasing ubiquity of digital devices, leveraging technology as an educational tool presents an engaging and impactful method for raising awareness about environmental issues.

Digital games, particularly serious games, have proven to be powerful tools in educational contexts. According to Tarouco, Konrath, and Grando (2004), "*Digital games can be effective instructional tools because they entertain while motivating, facilitate learning, and enhance the retention of what is taught by engaging the mental and intellectual functions of the player*". Within this framework, persuasive games offer unique opportunities to influence behavior and decision-making by immersing players in simulated environments where they can experiment and learn through direct interaction.

Recognizing the potential of digital games in environmental education, this study introduces "BORA CATÁ," a persuasive game developed to raise awareness about selective waste collection. By integrating visual, auditory, and interactive elements, the game aims to educate players on proper waste segregation practices in an engaging and effective manner.

1.1 OBJECTIVES

The objectives of this study are as follows:

Primary Objective: To develop and evaluate a digital educational game, "BORA CATÁ," aimed at promoting awareness and encouraging behavior change regarding selective waste collection and solid waste management.

Secondary Objectives:

Identify and implement the key educational content and game mechanics related to waste segregation.

Collect user feedback on the game's clarity, usability, and effectiveness in conveying environmental concepts.

Assess player satisfaction and engagement through gameplay.

Measure the game's impact on participants' understanding of waste management practices and its potential to foster sustainable habits.

By focusing on these objectives, this study seeks to explore how digital tools, specifically serious games, can contribute to environmental education and promote actionable change in the context of sustainable development.

2 THEORETICAL FRAMEWORK

The management of solid waste has become a pressing global challenge, demanding innovative and effective solutions to foster environmental awareness and sustainable practices. Theoretical discussions in the literature highlight the importance of integrating environmental education with emerging technologies to address these challenges. The concepts of serious games, persuasive games, and digital tools in educational contexts form the foundation of this study.

2.1 ENVIRONMENTAL EDUCATION AND WASTE MANAGEMENT

Environmental education plays a pivotal role in raising awareness about sustainable waste management. As highlighted by Silva et al. (2002), effective education should "sensitize individuals to reduce consumption, reuse and recycle waste, and rethink attitudes that degrade the environment." The incorporation of waste management topics into educational curricula encourages critical thinking and fosters a sense of responsibility in individuals, particularly when introduced during childhood.

2.2 SERIOUS GAMES AND PERSUASIVE GAMES

Serious games are characterized by their ability to combine entertainment with education, delivering meaningful content through an engaging medium. According to Xexéo et al. (2017), serious games are designed not only to entertain but also to "transmit messages, teach lessons, and provide experiences for players." These games leverage interactivity and immersion to enhance cognitive and emotional engagement, thereby supporting behavior change and learning.

Persuasive games, a subset of serious games, are particularly relevant in this context. They aim to influence players' decision-making and attitudes by embedding meaningful scenarios and challenges. As described in the dissertation, persuasive games can be "a fresh alternative compared to conventional persuasion techniques," as they immerse players in interactive experiences that encourage reflection and action beyond the virtual environment.

2.3 TECHNOLOGY AS A TOOL FOR ENVIRONMENTAL AWARENESS

Digital technologies have significantly transformed how educational content is delivered and absorbed. The proliferation of devices such as computers, tablets, and smartphones offers new opportunities to engage users in learning about environmental issues. Tarouco, Konrath, and Grando (2004) assert that digital games are effective instructional tools because they "entertain while motivating, facilitate learning, and enhance the retention of what is taught." In this regard, the integration of digital tools into environmental education has shown promise in promoting sustainable habits.

2.4 COMMITMENT TO THE SUSTAINABLE DEVELOPMENT GOALS

The development of the "BORA CATÁ" game aligns closely with the United Nations Sustainable Development Goals (SDGs), specifically Goals 4, 12, 13, 14, and 15 (Figura 1). These goals emphasize quality education, responsible consumption and production, climate action, the conservation of life below water, and the sustainable use of terrestrial ecosystems.

Figura 1

United Nations Sustainable Development Goals (SDGs).



Source: United Nations. (2023).

By addressing Goal 4, the game promotes lifelong learning opportunities through interactive environmental education. Goal 12 is reflected in its focus on waste segregation and recycling, encouraging responsible consumption patterns. Goal 13 is supported by fostering awareness and actions that mitigate the impacts of climate change through waste management practices. Finally, the game highlights the importance of protecting aquatic and terrestrial ecosystems (Goals 14 and 15), showcasing the interconnectedness of proper waste disposal and environmental conservation.

This integration demonstrates the broader impact of educational tools like "BORA CATÁ" in fostering global sustainability and supports the UN's call for inclusive and actionable approaches to achieving these objectives by 2030.

2.5 GAPS AND OPPORTUNITIES IN THE LITERATURE

Despite the growing interest in serious games and their application in education, gaps remain in their use for environmental awareness, particularly in selective waste collection. While previous studies have demonstrated the efficacy of games in motivating behavior change, few have focused specifically on waste management as a central theme. Additionally, there is a lack of research addressing the long-term impact of such games on players' habits and attitudes toward sustainability.

This study seeks to address these gaps by developing and evaluating "BORA CATÁ," a persuasive game designed to educate and engage users in the practice of selective waste collection. The research builds on existing theories and models while contributing to the growing body of literature on the intersection of environmental education and digital technologies.

3 METHODOLOGY

3.1 TYPE OF STUDY

This study employed a design and development research approach to create and evaluate "BORA CATÁ," a persuasive digital game aimed at raising awareness about selective waste collection. The research integrates theoretical and applied methodologies, aligning with the principles of serious games and persuasive learning strategies.

3.2 SAMPLE SELECTION

Participants were selected using a convenience sampling method. The sample consisted of individuals aged 18 to 30, the primary target demographic for the study, given their role as active decision-makers in society and their familiarity with digital technologies. A total of 4,969 individuals participated in the game evaluation phase, including 925 children, highlighting the broader reach of the game beyond the intended audience.

3.3 DATA COLLECTION METHODS

Data collection involved both qualitative and quantitative approaches:

- Game Interaction: Participants engaged with the "BORA CATÁ" game, which was developed using Unity and designed to simulate scenarios requiring proper waste segregation. The game rewarded or penalized players based on their actions.
- Surveys and Questionnaires: Post-gameplay surveys collected feedback on various aspects of the game, including usability, engagement, educational effectiveness, and overall satisfaction. Questions assessed participants' perceptions of the game's clarity, challenge, and ability to promote sustainable behaviors.

- Behavioral Metrics: Metrics such as in-game actions, decision-making patterns, and time spent on tasks were analyzed to assess engagement and learning outcomes.

3.4 DATA ANALYSIS

Data were analyzed using descriptive and inferential statistics to identify patterns in participants' feedback and gameplay behaviors. The qualitative data, derived from open-ended survey questions, were coded thematically to uncover insights regarding participants' perceptions and suggestions for improvement. Quantitative analysis focused on comparing pre- and post-intervention awareness levels regarding selective waste collection.

3.5 ETHICAL CONSIDERATIONS

The study adhered to ethical research principles, ensuring voluntary participation, anonymity, and confidentiality of all participants. Participants were informed about the study's objectives and provided informed consent before engaging with the game or surveys.

4 RESULTS AND DISCUSSIONS

The results obtained from the development and testing of the BORA CATÁ game highlight its effectiveness in promoting environmental awareness and selective waste collection. The game was tested by a diverse group of participants, and the data collected indicate a positive reception of the product.

Initially, the game demonstrated a user-friendly and intuitive interface, with the educational mechanics well integrated into the narrative. The developed graphs and tables, such as those related to usage frequency and the game's aesthetic perception, show a high approval rate regarding design and functionality. For instance, 90% of participants reported that the game facilitated learning about selective waste collection, while 85% indicated that the visual design significantly contributed to the user experience.

Figure 2

The main screen of BORA CATÁ, showcasing its intuitive and colorful design.



Source: Souza (2023).

As noted by Souza (2023), the player experience was carefully designed to balance educational content with engaging gameplay mechanics. The game provided players with a series of interactive challenges that simulated real-life scenarios of waste segregation. Players were tasked with identifying and categorizing different types of waste, receiving immediate feedback on their actions. This approach not only reinforced learning but also motivated players to reflect on their habits and behaviors regarding waste management. For instance, Phase 1 (Figure 3) introduces players to the basic mechanics of selective waste segregation through a backyard scenario. This phase allows users to interact with different types of waste using drag-and-drop mechanics. The design encourages players to intuitively identify recyclable and non-recyclable materials, laying the foundation for more complex challenges in subsequent phases.

Figure 3

Screenshot of Phase 1 of BORA CATÁ, illustrating the game's introductory challenges.



source: Souza (2023).

Participants reported that the immersive and interactive nature of the game enhanced their engagement and enjoyment. Features such as auditory cues, visual aids, and the gamified scoring system contributed to a positive user experience. Many players expressed that the game made the subject of waste segregation more relatable and easier to understand. Specific game elements, such as dynamic difficulty adjustment and reward systems, were particularly effective in maintaining player interest and encouraging repeated gameplay.

Figure 4

Arrangement of waste in the third and final phase of BORA CATÁ.



Source: Souza (2023).

In the discussion, comparisons with existing literature indicate that BORA CATÁ aligns with modern trends in educational and persuasive games. Studies like those by McGonigal (2011) and Santana (2008) reinforce that digital games are effective tools for engagement and education, particularly on sustainability-related topics. The use of digital technologies was highlighted as a bridge to reach both young people and adults, promoting behavioral changes in waste disposal.

However, limitations include the game's dependence on the Windows operating system, which restricts accessibility for users of other devices. Additionally, participants suggested improvements such as the inclusion of more levels and challenges to enhance the educational impact.

Finally, the results of this study significantly contribute to the field of digital environmental education, suggesting that interactive tools like BORA CATÁ can transform educational practices and promote more sustainable behaviors. It is recommended that the

project continues, exploring other platforms and expanding the scope to include advanced waste management practices.

Finally, the results of this study significantly contribute to the field of digital environmental education, suggesting that interactive tools like BORA CATÁ can transform educational practices and promote more sustainable behaviors. It is recommended that the project continues, exploring other platforms and expanding the scope to include advanced waste management practices.

5 CONCLUSION

he development of the digital game "BORA CATÁ" represents a significant contribution to environmental education, particularly in promoting awareness about selective waste collection. The game effectively combines elements of entertainment and education, leveraging interactive digital tools to engage users and influence their behaviors. Through immersive gameplay, players are encouraged to adopt sustainable practices, aligning their actions with the principles of environmental responsibility.

The study achieved its objectives by designing and evaluating a persuasive game that emphasizes the importance of proper waste segregation. Feedback from participants demonstrated the game's effectiveness in raising awareness, enhancing understanding, and fostering a sense of responsibility towards environmental preservation. The results highlighted positive reception and engagement, particularly among younger players, underscoring the potential of digital games as educational tools.

This research also reinforces the alignment of "BORA CATÁ" with the United Nations Sustainable Development Goals (SDGs), specifically Goals 4, 12, 13, 14, and 15. By addressing issues such as quality education, responsible consumption, and environmental conservation, the game contributes to global efforts towards sustainable development.

The developed software is available for public access and use at https://cjannuzzi.github.io/bora-cata/. This link allows individuals and institutions to explore and utilize the game as an educational tool, further amplifying its impact on raising environmental awareness and encouraging sustainable practices.

Despite its successes, the study identified limitations, including the need for broader demographic reach and long-term impact assessment. Future research could explore the scalability of such initiatives, integrating additional features to enhance user experience and measure behavioral changes over extended periods.

In conclusion, "BORA CATÁ" exemplifies the transformative potential of digital tools in addressing global environmental challenges. By fostering education and engagement, the game serves as a model for integrating technology into strategies for promoting sustainability and ecological awareness. This work not only advances the field of serious games but also underscores their relevance in contemporary environmental education.

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